SAFETY DATA SHEET

Date Prepared : 11/05/2015
SDS No : ALC-13-011-F

BASIC BLUE SF

1. PRODUCT AND COMPANY IDENTIFICATION

PRODUCT NAME: BASIC BLUE SF
PRODUCT DESCRIPTION: Proprietary Alkaline Cleaning Formulation

SUPPLIER
Broadmoor Products, Inc.
4201 Brockton Drive, S.E.
Grand Rapids, MI 49512
Customer Service: (616) 285-6440

24 HR. EMERGENCY TELEPHONE NUMBER
P.E.R.S. (US Transportation/ Spill): (800) 633-8253
For after-hours transportation and spill emergencies only.
Direct all other calls to the Customer Service number at left.

2. HAZARDS IDENTIFICATION

GHS CLASSIFICATIONS
Health:
Serious Eye Damage, Category 1
Skin Corrosion, Category 1C
Environmental:
Acute Hazards to the Aquatic Environment, Category 3
Chronic Hazards to the Aquatic Environment, Category 3

GHS LABEL


Corrosion

SIGNAL WORD: DANGER

HAZARD STATEMENTS
H314: Causes severe skin burns and eye damage.
H412: Harmful to aquatic life with long lasting effects.

PRECAUTIONARY STATEMENT(S)
Prevention:
P280: Wear protective gloves/protective clothing/eye protection/face protection.
P264: Wash hands /any other skin areas possibly contacted by the material thoroughly with soap and water after handling.
P260: Do not breathe mist/spray.
P273: Avoid release to the environment.
Response:
P305+P351+P338: IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P303+P361+P353: IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/ shower.
P363: Wash contaminated clothing before reuse.
P304+P340: IF INHALED: Remove person to fresh air and keep comfortable for breathing.
P301+P330+P331: IF SWALLOWED: Rinse mouth. Do NOT induce vomiting.
P310: Immediately call a POISON CENTER, doctor, or for EMERGENCY MEDICAL ASSISTANCE after initiating the first aid measures listed above.

P321: Specific treatment (see First Aid information on label or Section 4 of the SDS).

Storage:

P405: Store locked up.

Disposal:

P501: Dispose of contents/container in accordance with all applicable local, state, and federal regulations. (See Section 13 of the SDS). Be aware that residues in empty containers remain acutely hazardous.

EMERGENCY OVERVIEW

PHYSICAL APPEARANCE: Clear blue liquid

IMMEDIATE CONCERNS:

OSHA HAZARDS

Corrosive. Causes burns or severe irritation to eyes, skin, and mucous membranes.

POTENTIAL HEALTH EFFECTS

EYES: Corrosive to eye tissues. Causes severe irritation, inflammation and chemical burns, with possible permanent eye injury or blindness if not immediately and completely removed.

SKIN: Corrosive or severely irritating to skin. Contact causes defatting and drying of the skin, leading to irritation and cracking, with possible chemical burns if not promptly removed.

SKIN ABSORPTION: No skin absorption toxicity effects are reported in the health and safety literature for any component present at ≥ 1%.

INGESTION: Corrosive to the tissues of the gastrointestinal tract. Causes severe irritation and gastrointestinal symptoms such as pain, nausea, and vomiting if swallowed. May cause chemical burns to the mouth, throat, and stomach.

INHALATION: Inhalation of mists and sprays of concentrated product solutions, or dusts of dried material, may cause irritation (possibly severe) of respiratory tract tissues

PHYSICAL HAZARDS: Concentrated product may slowly corrode or etch active metals such as aluminum, tin, zinc, and galvanized materials, with the formation of limited quantities of flammable hydrogen gas. May react vigorously with concentrated strong acids.

COMMENTS: See Section 11: TOXICOLOGICAL INFORMATION.

Hazards Not Otherwise Classified: None known.

3. COMPOSITION / INFORMATION ON INGREDIENTS

Chemical Name	Wt.%	CAS
Sodium Hydroxide	< 1	1310-73-2
Tetrapotassium Pyrophosphate	< 2	7320-34-5
surfactant blend	6 - 10	proprietary
Water	> 85	7732-18-5

COMMENTS: While the identity of some components is claimed as a trade secret in accordance with the provision of OSHA 29 CFR 1910.1200(i), all known hazards are clearly communicated within this document. Exact percentages of the components of this proprietary formulation are considered confidential. Any additional unlisted components of this proprietary product formulation are either considered non-hazardous or are present at concentrations below regulated levels.

4. FIRST AID MEASURES

EYES: Immediately flush eyes with a directed stream of water for at least 20 minutes, forcibly holding eyelids apart to ensure complete irrigation of all eye and eyelid tissues. Remove contact lenses, if present, after the first minute of flushing. "Roll" eyes to fully expose eye surfaces to irrigation. GET IMMEDIATE MEDICAL ATTENTION.

SKIN: Immediately flush contaminated areas with water for at least 15 minutes. Remove all contaminated clothing, jewelry, and shoes immediately, taking care not to contaminate eyes. Thoroughly wash and dry contaminated clothing before re-use. SEEK IMMEDIATE MEDICAL ATTENTION if large areas of skin were exposed, or if burns, rash, or persistent irritation develop.

INGESTION: If swallowed, DO NOT induce vomiting. Never give anything by mouth to a person who is unconscious, convulsing, or unable to swallow. If victim is conscious and able to swallow, immediately rinse out mouth with water, and then give one cup (8 ounces) of milk or water to drink. To prevent aspiration of swallowed material, lay victim on side with head lower than waist. If vomiting occurs spontaneously, keep airway clear. Do not leave victim unattended. GET MEDICAL ATTENTION IMMEDIATELY.

INHALATION: If inhalation exposure to misted material occurs, or if symptoms develop, move to fresh air. If symptoms are severe or persistent, seek prompt medical attention. If breathing is difficult, oxygen should be administered by qualified personnel. If respiration or pulse has stopped, have a trained person administer basic life support (cardiopulmonary resuscitation with rescuer protection and/or Automatic External Defibrillator) as soon as possible, and CALL FOR EMERGENCY MEDICAL SERVICES IMMEDIATELY.

SIGNS AND SYMPTOMS OF OVEREXPOSURE

EYES: Immediate effects may include severe irritation, tearing, swelling, reddening, and blurred vision. If not promptly and completely removed, may cause corneal injury, tissue damage, severe chemical burns, and permanent eye damage.

SKIN: May include slippery feeling of skin, followed by irritation, drying and cracking, and dermatitis. Prolonged untreated contact causes burns, ulceration, and possible permanent tissue damage.

INGESTION: May include nausea, vomiting, diarrhea, gastrointestinal upset and/or ulceration. May result in difficulty in swallowing and breathing, and burns of the mouth and throat.

INHALATION: Inhalation of mists and sprays of concentrated product solutions, or dusts of dried material, may cause irritation of the respiratory tract, with upper respiratory tract symptoms such as runny nose and sore throat, coughing, and in severe overexposures, possibly mild burns to the mucous membranes.

CHRONIC EFFECTS: None known

NOTES TO PHYSICIAN: The absence of visible signs or symptoms of burns does not reliably exclude the presence of actual tissue damage. Possible mucosal damage may contraindicate the use of gastric lavage.

ANTIDOTES: No specific antidote.

ADDITIONAL INFORMATION: Ensure that emergency personnel are aware of the material(s) involved and take precautions to protect themselves.

COMMENTS: Refer to Section 2 (Hazards) and Section 11 (Toxicological Information) for additional information on symptoms of overexposure.

5. FIRE FIGHTING MEASURES

FLAMMABLE CLASS: Not combustible unless evaporated to dryness.

EXTINGUISHING MEDIA: Use water, carbon dioxide, foam, or dry chemical as appropriate to extinguish surrounding fire.

EXPLOSION HAZARDS: Not applicable unless there has been contact with active metals causing the formation of hydrogen gas

FIRE FIGHTING PROCEDURES: Material is harmful to aquatic organisms. Prevent fire run-off from entering drains, storm sewers, and natural waterways to prevent damage to the environment.

FIRE FIGHTING EQUIPMENT: Wear full turn-out gear and SCBA with full face protection when fighting fires involving this material.

HAZARDOUS DECOMPOSITION PRODUCTS: Ignition of evaporated residues produces fumes and smoke containing oxides of carbon, nitrogen, sulfur, and phosphorus.

6. ACCIDENTAL RELEASE MEASURES

SMALL SPILL: Wear appropriate Personal Protective Equipment (as described in Section 8 of the SDS) to prevent any contact with eyes, skin, and clothing. Avoid stepping in material, which is slippery and poses a slip-and-fall hazard. Ensure adequate ventilation. Absorb spill with suitable absorbent (e.g., sand, pet litter, universal absorbent, diatomaceous earth, etc.) Collect in suitable and properly labeled containers for proper disposal. Or neutralize spill with a mild acid (such as acetic acid) or sodium bicarbonate, and mop up. Flush remaining trace residues to sanitary sewer with plenty of water.

LARGE SPILL: Wear appropriate personal protective equipment (as described in Section 8 of the SDS) to prevent any contact with eyes, skin, and clothing. Provide adequate ventilation or wear properly fitted respiratory protection equipment. CAUTION! Spilled material produces extremely slippery surfaces. Completely contain spilled material with absorbent socks, dikes of sand or earth, etc., to prevent entry into drains, storm sewers, and natural waterways. Consider use of a suitable vacuum or pump to collect as much material as possible in a suitable clean container for use as intended. Use sand, universal absorbent, or similar material to absorb the spill and shovel into closable containers for proper disposal. Remaining residues can be neutralized with a dilute acid (such as acetic acid) or sodium bicarbonate, and flushed to sanitary sewer with large quantities of water.

7. HANDLING AND STORAGE

GENERAL PROCEDURES: Always wear the recommended Personal Protective Equipment when handling this material to avoid any contact with eyes, skin, and clothing. (See Section 8)

HANDLING: Handle and open container with care. Follow all SDS/label precautions even after container is emptied because of retained product residues.

STORAGE: Store in a secured area out of the reach of children. Keep containers tightly closed when not in use. Store away from acids and other incompatible materials.

8. EXPOSURE CONTROLS / PERSONAL PROTECTION

EXPOSURE GUIDELINES

OSHA HAZARDOUS COMPONENTS (29 CFR1910.1200)				
	EXPOSURE LIMITS			
Chemical Name	**Type**		**ppm**	**mg/m³**
Sodium Hydroxide	**OSHA PEL**	**TWA**		2
	ACGIH TLV	**STEL**		2 (ceiling)
Tetrapotassium Pyrophosphate	**OSHA PEL**	**TWA**	Not Listed [1]	[1]
		STEL	Not Listed	
	ACGIH TLV	**TWA**	Not Listed [2]	[2]
		STEL	Not Listed	
surfactant blend	**OSHA PEL**	**TWA**	Not Listed	
		STEL	Not Listed	
	ACGIH TLV	**TWA**	Not Listed	
		STEL	Not Listed	

Footnotes:
1. OSHA PELs = 15 mg/m^3 total and 5 mg/m^3 respirable fraction of particulates not otherwise regulated ("Nuisance Dust").
2. ACGIH TLVs = 10 mg/m^3 inhalable particulate and 3 mg/m^3 respirable fraction of particulates not otherwise specified ("Nuisance Dust").

ENGINEERING CONTROLS: Provide adequate ventilation. Provide local exhaust when high concentrations of mists or aerosols are formed by spray applications. If ventilation is inadequate to maintain airborne concentrations below exposure limits, use properly fitted NIOSH-approved full-face respiratory equipment for protection against mists/ dusts.

PERSONAL PROTECTIVE EQUIPMENT

EYES AND FACE: Wear splash-proof chemical goggles. Do not wear contact lenses. Use of a full face shield to protect the face and head is recommended whenever engaged in operations that may result in splashing or splattering of the concentrated product.

SKIN: Wear appropriate impervious chemical resistant gloves. (Viton, natural rubber, neoprene, nitrile, PVC, polyethylene, and ethyl vinyl alcohol laminate are suitable materials.)

RESPIRATORY: Use of respiratory protection is usually not required. However, if use conditions generate mists, aerosols, or dusts, and ventilation is inadequate, or where symptoms of overexposure are occurring, wear properly fitted NIOSH-approved respiratory equipment for protection against mists and dusts.

PROTECTIVE CLOTHING: Wear impervious chemical resistant apron or suit and rubber boots when potential for bodily contact with the concentrated product exists. Contaminated clothing should be immediately removed and laundered.

WORK HYGIENIC PRACTICES: Always use the recommended personal protective equipment to prevent any contact with eyes and skin. Wash skin and contaminated clothing thoroughly after handling. Do not eat, drink or smoke when using this product. Handle in accordance with good industrial hygiene and safety practices.

OTHER USE PRECAUTIONS: A safety shower and eye wash fountain or adequate source of water should be located in the immediate vicinity.

9. PHYSICAL AND CHEMICAL PROPERTIES

PHYSICAL STATE: Liquid
ODOR: Somewhat pungent detergent odor
ODOR THRESHOLD: Not available

APPEARANCE: Clear blue liquid
pH: 12.5 to 13.3 (neat liquid)
FLASH POINT AND METHOD: >200°F (Not flammable or combustible unless evaporated to dryness.)
FLAMMABLE LIMITS: NA = Not Applicable
AUTOIGNITION TEMPERATURE: NA = Not Applicable
VAPOR PRESSURE: <18 mm Hg (2.4 kPa) at 68°F / 20°C
VAPOR DENSITY: Approximately the same as air
BOILING POINT: ≥212°F (100°C)
FREEZING POINT: 31°F - 32°F (-0.5°C to 0°C)
THERMAL DECOMPOSITION TEMPERATURE: Not Established
SOLUBILITY IN WATER: Complete
PARTITION COEFFICIENT: N-OCTANOL/WATER: Not Established
EVAPORATION RATE: Approximately the same as water.
DENSITY: Slightly more dense than pure water.
SPECIFIC GRAVITY: 1.01 - 1.03
VISCOSITY: Not Established

10. STABILITY AND REACTIVITY

REACTIVITY: No
HAZARDOUS POLYMERIZATION: No
STABILITY: Stable under normal storage and use conditions.
POSSIBILITY OF HAZARDOUS REACTIONS: May react vigorously with concentrated strong acids. Concentrated product may etch or slowly corrode reactive metals (such as aluminum, tin, zinc, and galvanized metals).
HAZARDOUS DECOMPOSITION PRODUCTS: Ignition of evaporated residues produces fumes and smoke containing oxides of carbon, nitrogen, sulfur, and phosphorus.
INCOMPATIBLE MATERIALS: Acids; Halogenated compounds. Concentrated product may be corrosive toward active metals such as aluminum, tin, zinc, and galvanized materials, especially when heated, with the formation of flammable hydrogen gas. Concentrated product may etch or slowly corrode brass, bronze, copper, and copper alloys.

11. TOXICOLOGICAL INFORMATION

ACUTE TOXICITY

DERMAL LD_{50}: No testing data are available for the product as a whole. Acute dermal harmful effects are primarily due to the aggressive corrosive and defatting effects of the caustic alkalinity coupled with strong detergency.

ORAL LD_{50}: No testing data are available for the product as a whole. Harmful acute oral effects are primarily due to alkaline corrosivity; all other components exhibit comparatively lower levels of acute oral toxicity at the levels present in the product..

INHALATION LC_{50}: No testing data are available for the product as a whole. Harmful acute inhalation effects of mists or dusts of dried material are primarily due to alkaline corrosivity.

Notes: Immediately Dangerous to Life and Health (IDLH) value for inhalation exposure to Sodium Hydroxide = 10 mg/m^3.

NOTES: Acutely hazardous due to corrosivity or strong irritantancy on all tissues of the eye, skin, and mucous membranes. The presence of surfactants is likely to accelerate the rate of corrosive action of sodium hydroxide on tissues.

RESPIRATORY OR SKIN SENSITIZATION: No reports of sensitization have been found in the health and safety literature for any of the individual components present at ≥ 0.1%.

GERM CELL MUTAGENICITY: No reports of mutagenicity have been found in the health and safety literature for any of the components present at ≥ 0.1%.

CARCINOGENICITY

IARC: No component of this product present at levels greater than or equal to 0.1% is listed as a probable or confirmed human carcinogen by the International Agency for Research on Cancer.
Contains low levels (<0.04%) trisodium nitrilotriacetate (CAS 5064-31-3) that is listed by IARC as Category 2B, "Possibly Carcinogenic to Humans," based on animal testing in which massive dietary doses produced urinary tract tumors. However, especially at subtoxic doses, there is little likelihood this compound could cause cancer in humans. The product may also contain <0.002% formaldehyde (CAS 50-00-0), a known carcinogen.

NTP: No component of this product present at levels greater than or equal to 0.1% is listed as a probable or confirmed human carcinogen by the National Toxicology Program.
May contain trace levels of Trisodium Nitrilotriacetate that is listed by NTP as Category R, "Reasonably Anticipated to be a Human Carcinogen," and formaldehyde, that is is listed by NTP as Category K, "Known to be a Human Carcinogen."

OSHA: No component of this product present at levels greater than or equal to 0.1% is listed as a probable or confirmed human carcinogen by OSHA.

REPRODUCTIVE TOXICITY: None reported from laboratory studies on the individual components at doses below maternally toxic levels.

GENERAL COMMENTS: No data for other effects are available.

12. ECOLOGICAL INFORMATION

ECOTOXICOLOGICAL INFORMATION: This material exhibits moderately severe acute toxicity to aquatic organisms. Harmful effects on the aquatic environment are expected to result primarily from surfactant toxicity, with additional harmful effects from pH elevation.

BIOACCUMULATION/ACCUMULATION: Little potential for significant bioaccumulation is anticipated due to the high water solubility of the ingredients.

AQUATIC TOXICITY (ACUTE): No testing data are available for the product as a whole. LC_{50} values for freshwater fish (96 hr) and EC_{50} values for daphnia (48 hr) are estimated to in the range of 10 to 30 mg/L, based on weighted averages of literature toxicity values for the individual components.

GENERAL COMMENTS: Direct discharges into the aquatic environment are prohibited and must be prevented. Do not discharge the product into drains, storm sewers, or natural waterways. Material at use dilutions is effectively removed by conventional treatment in wastewater plants.

COMMENTS: Contains low levels of phosphate that may contribute to the eutrophication of natural waters. (Phosphorus content <0.7% as PO_4^{3-}).

13. DISPOSAL CONSIDERATIONS

PRODUCT DISPOSAL: Dispose of only in conformity with all applicable federal, state, and local regulations.

EMPTY CONTAINER: Be aware that product residues in empty containers remain acutely hazardous.

RCRA/EPA WASTE INFORMATION: For disposal, material is considered hazardous waste with RCRA characteristic of corrosivity (D002), as defined by 40CFR §262.22.

14. TRANSPORT INFORMATION

DOT (DEPARTMENT OF TRANSPORTATION)

PROPER SHIPPING NAME: UN1760, Corrosive Liquids, N.O.S. (contains Sodium Hydroxide, Tetrapotassium Pyrophosphate), 8, PGIII

NAERG: 154

REPORTABLE QUANTITY (RQ) UNDER CERCLA: Not applicable to non-bulk packages (less than 10,000 gallons) of this product.

MARINE POLLUTANTS: None

15. REGULATORY INFORMATION

UNITED STATES

SARA TITLE III (SUPERFUND AMENDMENTS AND REAUTHORIZATION ACT)

FIRE: No **PRESSURE GENERATING:** No **REACTIVITY:** No **ACUTE:** Yes **CHRONIC:** No

302/304 EMERGENCY PLANNING

THRESHOLD PLANNING QUANTITY / REPORTABLE QUANTITIES: No listed components.

CERCLA (COMPREHENSIVE RESPONSE, COMPENSATION, AND LIABILITY ACT)

CERCLA RQ: Not applicable to non-bulk packages (less than 10,000 gallons) of this product.

TSCA (TOXIC SUBSTANCE CONTROL ACT)

TSCA STATUS: All components of this product are listed on the Toxic Substance Control Act (TSCA) inventory, or are exempt.

16. OTHER INFORMATION

REASON FOR ISSUE: New SDS format (GHS compliant). This version replaces all previous versions.

APPROVED BY: Technical Director

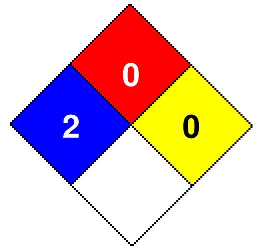
PREPARED BY: Technical Services - Broadmoor Products **Date Prepared:** 11/05/2015

INFORMATION CONTACT: Technical Services - Broadmoor Products, Inc.: (616) 285-6440

HMIS RATING

HEALTH	2
FLAMMABILITY	0
PHYSICAL HAZARD	0
PERSONAL PROTECTION	C

NFPA CODES



GENERAL STATEMENTS: Each user of this product is expected and urged to read and understand the entire SDS (consulting appropriate expertise as necessary) in order to comprehend the information contained in this SDS and any hazards associated with the product, with the expectation that the precautions identified in this document will be followed unless your use conditions would necessitate other appropriate methods or actions.

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